

GES 212.2

Philosophy, Logic, and Critical Thinking

Study Notes — Godswill | UNIPORT, General Studies

Chapter Overview

This course covers the foundations of philosophical thinking, moving from what philosophy is and where it came from, through its two great theoretical branches — metaphysics (the study of reality) and epistemology (the study of knowledge) — into the tools philosophy gives us to reason well: logic, critical thinking, argument forms, and symbolic logic. It then turns to philosophy's practical/applied side: ethics (how we should live and treat one another), philosophy of religion (God, the soul, and ultimate questions), political philosophy (the state, justice, and obligation), and finally the existentialist tradition's questions about freedom, authenticity, anxiety, and death — philosophy's most direct engagement with the meaning of human existence itself.

Unit One: The Meaning, Nature, Branches, and Scope of Philosophy

1.1 What Is Philosophy?

The word 'philosophy' comes from the Greek *philos* (love) and *sophia* (wisdom) — literally, 'the love of wisdom.' Philosophy is the systematic, critical study of fundamental questions about reality, knowledge, existence, reason, mind, and values. Rather than accepting claims on authority or tradition alone, philosophy insists on examining the reasons and evidence behind a belief.

The Greek thinker Pythagoras is traditionally credited with coining the term: he reportedly described himself not as a *sophos* (a wise man) but as a *philosophos* — a lover of wisdom — reflecting the idea that wisdom is something to be pursued rather than fully possessed.

1.2 The Nature of Philosophy

Philosophy is characterized by critical thinking, rational inquiry, and systematic analysis. It differs from religion in that religion is often grounded in faith and revelation, while philosophy is grounded in reason and argument. It differs from science in that science investigates the natural world through observation and experiment to establish empirical facts, while philosophy asks more fundamental, often normative or conceptual questions — including questions about the assumptions science itself relies on.

1.3 The Major Branches of Philosophy

Branch	What It Studies
Metaphysics	The fundamental nature of reality and being — what exists, and what it means to exist.
Epistemology	The nature, sources, and limits of knowledge — what we can know, and how.
Ethics	Moral principles and values — how we ought to act and live.
Logic	The principles of correct/valid reasoning and argument.
Aesthetics	The nature of beauty, art, and taste.
Political Philosophy	The state, justice, authority, and political obligation.

Branch	What It Studies
Philosophy of Religion	The rational examination of religious concepts — God, the soul, and religious experience.

1.4 The Scope and Foundational Problems of Philosophy

Because philosophy questions the fundamental assumptions underlying every field, its scope is unusually broad — it engages with science, ethics, politics, religion, art, and language alike. Its foundational problems include: what is real (metaphysics), what can we know and how (epistemology), how should we live (ethics), and what counts as good reasoning (logic) — questions that recur, in different forms, throughout the rest of this course.

Unit Two: The History of Philosophy — Ancient, Medieval, Modern, and Contemporary

2.1 The Ancient Period

Western philosophy traditionally begins with the Pre-Socratics, who sought a rational (rather than mythological) explanation for the natural world. Thales of Miletus is often called the first Western philosopher for proposing that water was the arche (first principle) underlying all things. Pythagoras emphasized number as the fundamental structure of reality; Heraclitus argued that everything is in constant flux ('you cannot step into the same river twice'); Parmenides argued, by contrast, that true being is unchanging and unified.

Socrates shifted philosophy's focus toward ethics and self-knowledge, using a method of probing questions (the Socratic method) to expose contradictions in his interlocutors' beliefs — famously urging others to 'know thyself.' His student Plato developed the Theory of Forms, arguing that abstract, perfect Forms (such as Justice or Beauty) are more real than the changing physical objects that merely imitate them — illustrated by his Allegory of the Cave. Plato's own student, Aristotle, took a more empirical turn, grounding knowledge in observation of the natural world, developing formal logic (the syllogism), the theory of the Four Causes, and a virtue-based approach to ethics.

2.2 The Medieval Period

Medieval philosophy was largely defined by the effort to reconcile faith and reason within Christian and Islamic thought. Augustine of Hippo blended Platonism with Christian theology, framing philosophy as 'faith seeking understanding.' Anselm of Canterbury formulated the ontological argument for God's existence — reasoning from the very concept of God as 'that than which nothing greater can be conceived' to the conclusion that God must exist. Thomas Aquinas, the central figure of Scholasticism, synthesized Aristotelian philosophy with Christian doctrine and offered his famous 'Five Ways' — cosmological arguments for God's existence.

In the Islamic world, philosophers such as Al-Farabi, Avicenna (Ibn Sina), Al-Ghazali, and Averroes (Ibn Rushd) — the latter known as 'The Commentator' for his extensive commentaries on Aristotle — preserved, translated, and extended Greek philosophy, profoundly shaping the later European scholastic tradition.

2.3 The Modern Period

Modern philosophy is often said to begin with René Descartes, sometimes called the father of modern philosophy. Using a method of systematic doubt, Descartes arrived at the famous conclusion *cogito, ergo sum* ('I think, therefore I am') as an indubitable starting point for knowledge, and argued for a fundamental dualism between mind and body.

Descartes' rationalism — the view that reason, not sensory experience, is the primary source of knowledge — was shared by Baruch Spinoza and Gottfried Leibniz. Opposing this, British empiricists argued that knowledge comes primarily from sensory experience: John Locke described the mind at birth as a tabula rasa ('blank slate') filled in by experience; George Berkeley pushed empiricism toward idealism with the principle esse est percipi ('to be is to be perceived'); David Hume applied empiricism rigorously enough to arrive at deep skepticism about causation and induction.

Immanuel Kant sought to reconcile rationalism and empiricism, arguing that knowledge requires both sensory experience and innate structures of the mind that organize that experience. In political philosophy, this period also produced the social contract tradition — Thomas Hobbes, John Locke, and Jean-Jacques Rousseau — examining the legitimacy of political authority (see Unit Twelve).

2.4 The Contemporary Period

The 19th and 20th centuries saw the rise of existentialism, beginning with Søren Kierkegaard (often called the father of existentialism) and Friedrich Nietzsche (known for proclaiming 'God is dead' and developing the concept of the will to power), and continuing through Jean-Paul Sartre ('existence precedes essence') and Albert Camus (philosophy of the absurd). Alongside this, phenomenology (Edmund Husserl, Martin Heidegger) examined the structures of conscious experience; analytic philosophy (Bertrand Russell, Ludwig Wittgenstein, Karl Popper) emphasized logical clarity and the philosophy of science; and postmodernism (Michel Foucault) critically examined power, knowledge, and discourse in society.

Unit Three: Key Themes in Philosophy

Beyond individual philosophers, several recurring theoretical positions cut across the history of philosophy and resurface throughout epistemology, metaphysics, and science.

Theme	Core Idea	Associated Thinkers
Rationalism	Reason, not sensory experience, is the primary source of genuine knowledge.	Descartes, Spinoza, Leibniz
Empiricism	Sensory experience is the primary (or only) source of knowledge.	Locke, Berkeley, Hume
Idealism	Reality is fundamentally mental or mind-dependent.	Berkeley, Hegel (German Idealism)
Positivism	Only empirically verifiable (scientifically testable) claims are meaningful.	Auguste Comte; Logical Positivism (Vienna Circle)
Pragmatism	The truth or meaning of an idea is determined by its practical consequences.	William James, John Dewey, Charles Peirce

A related recurring debate concerns Determinism versus Free Will: determinism holds that every event, including human choices, is the necessary result of prior causes, while libertarian free will holds that at least some human actions are freely and genuinely chosen, not fully determined by prior causes.

Unit Four: Metaphysics — Being, Non-Being, Substance, and Universals

4.1 What Metaphysics Studies

Metaphysics is the branch of philosophy concerned with the fundamental nature of reality — what exists, and what it means for something to exist. Ontology, the study of being and existence as such, is often treated as metaphysics' core subfield: it asks not just what particular things exist, but what 'existence' itself fundamentally consists in.

4.2 Substance

A substance, in the philosophical sense, is that which exists independently and underlies the properties a thing has — the 'thing itself' as opposed to the qualities attached to it. Aristotle treated substance as the primary category of being. Later philosophers took strikingly different positions on how many substances exist:

- Descartes' Dualism: reality consists of two distinct kinds of substance — mind (thinking substance) and body (extended, physical substance).
- Spinoza's Monism: there is only a single substance, which Spinoza identifies with both God and Nature.
- Leibniz's Monads: reality is composed of countless simple, indivisible substances called monads, each a self-contained unit reflecting the whole universe from its own perspective.

4.3 The Problem of Universals

A universal is a general property or category (such as 'redness' or 'humanity') that can apply to many particular things at once. The Problem of Universals asks whether such universals exist independently of the particular things that instantiate them, or whether they are merely useful labels/concepts with no independent reality.

Position	Core Claim
Realism	Universals exist independently of particular things (e.g. Plato's theory of Forms).
Nominalism	Universals are merely names/labels we apply; only particular things truly exist (associated with William of Ockham).
Conceptualism	Universals exist, but only as concepts within the mind, not independently in the external world.

Unit Five: Epistemology — Knowledge, Justification, and Its Fallible Nature

5.1 What Epistemology Studies

Epistemology is the branch of philosophy concerned with the nature, sources, scope, and limits of knowledge — what can we know, how do we come to know it, and how confident can we be in our knowledge claims?

5.2 The Traditional Definition of Knowledge

Classically, knowledge has been defined as Justified True Belief (JTB): to know a proposition, a person must (1) believe it, (2) it must be true, and (3) the belief must be justified (supported by adequate reasons or evidence). Edmund Gettier famously challenged this definition with counterexamples (the 'Gettier Problem') showing cases

where someone has a justified true belief that nonetheless doesn't seem to count as genuine knowledge, due to a kind of epistemic luck.

5.3 Theories of Justification

- Foundationalism: some beliefs are 'basic' — self-justifying or directly justified by experience — and all other beliefs are justified by being built upon these basic beliefs.
- Coherentism: a belief is justified by fitting coherently within a larger, mutually supporting web of beliefs, rather than resting on any single foundational belief.
- Reliabilism: a belief is justified if it is produced by a cognitive process that reliably tends to produce true beliefs (e.g. perception, memory, valid reasoning).

5.4 Fallibilism and Skepticism

Fallibilism is the view that our beliefs — even well-justified ones — could in principle turn out to be mistaken; justification does not require absolute certainty. This stands in contrast to more demanding classical standards for knowledge. Skepticism goes further, questioning whether genuine knowledge is possible at all, or at least in some domain — as illustrated by Descartes' method of systematic doubt and Hume's skepticism about our justification for believing in cause and effect.

Unit Six: The Concept, Nature, and Scope of Logic

6.1 What Is Logic?

Logic is the branch of philosophy (and of mathematics) concerned with the study of correct reasoning — the principles that distinguish valid inferences from invalid ones. Logic asks not what people actually believe, but what conclusions are properly, validly supported by given premises.

6.2 Logic and Related Disciplines

- Logic vs Psychology: psychology studies how people actually think (as a descriptive, empirical matter); logic studies how people ought to reason if they want to reason correctly (a normative matter).
- Logic vs Rhetoric: rhetoric is the art of persuasion, which may or may not rely on genuinely valid reasoning; logic is concerned strictly with the validity of the reasoning itself, independent of whether it happens to be persuasive.

6.3 Deductive vs Inductive Logic

- Deductive Logic: reasoning where, if the premises are true and the argument is valid, the conclusion must be true — the conclusion follows with logical necessity.
- Inductive Logic: reasoning that moves from specific observations to general conclusions that are probable, but not guaranteed, even if the premises are true.

Logic underpins critical thinking (Unit Seven) by supplying the formal standards — validity, soundness, and the avoidance of fallacies — against which everyday reasoning and argument can be evaluated and improved.

Unit Seven: Critical Thinking — Propositions, Arguments, and Fallacies

7.1 What Is Critical Thinking?

Critical thinking is the disciplined practice of analyzing, evaluating, and constructing reasoning carefully and objectively, rather than accepting claims uncritically. It draws directly on logic's tools to assess whether a piece of reasoning is actually sound.

7.2 Propositions and Arguments

A proposition (or statement) is a declarative sentence that is either true or false. An argument is a set of propositions in which one or more (the premises) are offered as support for another (the conclusion).

- **Validity:** an argument is valid if the conclusion logically follows from the premises — that is, if the premises were true, the conclusion could not be false.
- **Soundness:** a deductive argument is sound if it is both valid and all of its premises are actually true. An argument can be valid without being sound (if a premise is false).

7.3 Deductive, Inductive, and Abductive Reasoning

Beyond the deductive/inductive distinction (Unit Six), abductive reasoning (inference to the best explanation) starts from an observation and infers the explanation that would, if true, best account for it — common in everyday reasoning and scientific hypothesis-formation, though not guaranteed to be true.

7.4 Fallacies

A fallacy is an error in reasoning that undermines the logical validity of an argument, often while still appearing persuasive.

Fallacy	Description
Ad Hominem	Attacking the person making an argument rather than addressing the argument itself.
Straw Man	Misrepresenting an opponent's argument in a weaker form, then refuting that weaker version.
False Dilemma	Presenting only two options as if they were the only possibilities, when more exist.
Slippery Slope	Arguing that a small first step will inevitably lead to a chain of extreme, undesirable outcomes, without justifying that chain.
Circular Reasoning	An argument whose conclusion is simply restated as (or assumed by) one of its own premises.
Hasty Generalization	Drawing a broad conclusion from a sample that is too small or unrepresentative.
Appeal to Authority	Treating a claim as true solely because an authority figure asserts it, without independent support.

Unit Eight: Argument, Forms, and Syllogisms

8.1 Categorical Propositions

Traditional (Aristotelian) logic classifies statements about categories of things into four standard forms:

Form	Name	Example Pattern
A	Universal Affirmative	All S are P
E	Universal Negative	No S are P
I	Particular Affirmative	Some S are P
O	Particular Negative	Some S are not P

8.2 The Categorical Syllogism

A categorical syllogism is a deductive argument made up of exactly three categorical propositions: a major premise, a minor premise, and a conclusion, involving exactly three terms — the major term, the minor term, and the middle term (which appears in both premises but not in the conclusion, and serves to logically connect the major and minor terms).

A syllogism's validity depends strictly on its logical form (which terms are distributed, how the middle term connects the premises), not on whether its premises happen to be true in the real world — a syllogism can be logically valid while containing a false premise, and it can also be invalid despite having a true conclusion by coincidence.

Venn diagrams provide a visual method for testing whether a given categorical syllogism is valid, by diagramming the relationships between the three terms' categories and checking whether the conclusion is fully represented by what the diagrammed premises establish.

Unit Nine: Symbolic Logic and Rules of Inference

9.1 Symbolic Notation

Symbolic (propositional) logic represents statements and their logical relationships using symbols rather than natural language, making arguments easier to formally analyze.

Symbol	Operation	Meaning
$\neg$ (or $\sim$)	Negation	NOT — reverses the truth value of a statement.
$\wedge$	Conjunction	AND — true only when both statements are true.
$\vee$	Disjunction	OR — true when at least one statement is true.
$\rightarrow$	Implication (Conditional)	IF...THEN — false only when the antecedent is true and the consequent is false.
$\leftrightarrow$	Biconditional	IF AND ONLY IF — true when both statements share the same truth value.

9.2 Common Rules of Inference

- Modus Ponens: from 'If P then Q' and 'P,' infer 'Q.'
- Modus Tollens: from 'If P then Q' and 'not Q,' infer 'not P.'

- Disjunctive Syllogism: from 'P or Q' and 'not P,' infer 'Q.'
- Hypothetical Syllogism: from 'If P then Q' and 'If Q then R,' infer 'If P then R.'
- Conjunction: from 'P' and 'Q' separately, infer 'P and Q.'
- Simplification: from 'P and Q,' infer 'P' (or 'Q') alone.
- Addition: from 'P,' infer 'P or Q' (for any Q).
- Constructive/Destructive Dilemma: forms that reason validly across two conditional statements joined by a disjunction, either affirming an antecedent or denying a consequent across both branches.

These rules echo the same underlying logical laws (including De Morgan's Laws — see the Boolean Algebra material in computing-related courses) that make it possible to prove an argument's conclusion follows necessarily from its premises, step by step.

Unit Ten: Moral Philosophy (Ethics)

10.1 The Meaning and Branches of Ethics

Ethics (moral philosophy) is the branch of philosophy concerned with what is right and wrong, good and bad, in human conduct. It is generally divided into three branches:

- Metaethics: examines the nature and meaning of moral language itself — what does it mean to call something 'good,' and are moral facts objective or relative?
- Normative Ethics: develops and evaluates theories/standards for determining right action (see the major theories below).
- Applied Ethics: applies ethical theory to specific practical issues, such as bioethics, business ethics, and environmental ethics.

10.2 Major Ethical Theories

Theory	Core Claim	Key Figure(s)
Deontology	Actions are right or wrong based on adherence to moral duties/rules, regardless of consequences.	Immanuel Kant
Consequentialism / Utilitarianism	The rightness of an action depends on its consequences — specifically, maximizing overall well-being/happiness.	Jeremy Bentham, John Stuart Mill
Virtue Ethics	Right action flows from good character (virtues) rather than rules or outcomes alone.	Aristotle
Natural Law Theory	Morality is grounded in a rational order inherent in nature/human nature.	Thomas Aquinas

10.3 Philosophy's Role in Character Molding

Beyond describing ethical theories, philosophy plays a practical role in shaping character by training habits of critical self-reflection, weighing competing values and obligations, and reasoning carefully through moral dilemmas rather than relying purely on custom or unreflective emotion.

Unit Eleven: Philosophy of Religion

11.1 Nature and Scope

Philosophy of religion is the philosophical (rather than theological) examination of religious concepts and claims — using reason and argument to critically analyze religious texts, beliefs, and experiences, rather than presupposing their truth. Central questions include the existence of God, the immortality of the soul, the origin of the universe, and humanity's place within it.

11.2 Arguments for the Existence of God

- The Ontological Argument (Anselm): reasons a priori from the very concept of God as the greatest conceivable being to the conclusion that such a being must exist.
- The Cosmological Argument (Aquinas): reasons from the existence of the universe/change/causation back to a First Cause or Unmoved Mover — God.
- The Teleological (Design) Argument: reasons from the apparent order, complexity, and purposefulness of the universe to the conclusion that it must have an intelligent designer.
- The Moral Argument: reasons from the existence of objective moral values to the conclusion that a moral lawgiver (God) must exist as their source.

11.3 The Problem of Evil

The Problem of Evil challenges belief in an all-powerful, all-knowing, and perfectly good God by asking how such a God could permit the existence of suffering and evil in the world — a central topic in philosophy of religion, generating various theodicies (philosophical defenses attempting to reconcile God's existence with the reality of evil).

11.4 The Immortality of the Soul

Philosophy of religion also examines whether the soul (or some form of personal identity) survives bodily death, and what philosophical arguments — as opposed to purely religious/scriptural claims — might support or challenge this belief.

Unit Twelve: Political Philosophy

12.1 The Central Problem of Political Philosophy

Political philosophy examines the legitimacy of political authority and obligation — why (if at all) individuals are obligated to obey the state, and what makes a political system or its authority just.

12.2 Social Contract Theory

Social contract theory argues that political authority and obligation arise from an (explicit or implicit) agreement among individuals to form and submit to a governing authority, typically in exchange for security, order, or the protection of rights.

- Thomas Hobbes (Leviathan): argued that without government, humans exist in a 'state of nature' that is a war of all against all, making life 'solitary, poor, nasty, brutish, and short' — individuals therefore consent to an absolute sovereign to escape this condition.

- John Locke: argued that individuals possess natural rights (to life, liberty, and property) even in the state of nature, and government exists by the consent of the governed specifically to protect those rights — legitimizing resistance if government fails to do so.
- Jean-Jacques Rousseau: argued that legitimate political authority arises from the 'general will' of the people collectively, rather than from any single ruler's authority.

12.3 Key Concepts in Political Philosophy

- State: a political organization with a monopoly on legitimate authority over a defined territory and population.
- Nation: a community bound by shared identity — often culture, language, or history — which may or may not align with a single state's borders.
- Sovereignty: the supreme, independent authority of a state to govern itself without external interference.
- Justice: the fair and equitable distribution of rights, duties, and resources within a political community.
- Civil Disobedience: the deliberate, public, and typically non-violent violation of a law in order to protest its injustice or effect political/social change.

Unit Thirteen: Philosophy and Human Existence

13.1 Freedom and Responsibility

Existentialist philosophy places human freedom at the center of what it means to exist. Jean-Paul Sartre argued that 'existence precedes essence' — humans are not born with a fixed nature or purpose, but must freely create their own meaning and identity through their choices, which is why Sartre described human beings as 'condemned to be free': freedom is inescapable, and with it comes full responsibility for one's choices.

13.2 Authenticity and Bad Faith

Authenticity, in existentialist thought, means living in accordance with one's own true values and freely-chosen commitments, rather than passively conforming to external social roles, expectations, or excuses. Its opposite, 'bad faith' (Sartre), describes the act of denying one's own freedom and responsibility by pretending one's choices are forced or predetermined.

13.3 Anxiety, Alienation, and the Absurd

- Anxiety (Angst): for Kierkegaard and Heidegger, anxiety arises specifically from confronting the weight of one's own freedom and the open, undetermined possibilities of existence — distinct from ordinary fear, which has a specific object.
- Alienation: the feeling of isolation or estrangement — from oneself, from others, or from a meaningless world — a recurring theme in existentialist and social philosophy.
- The Absurd (Camus): the fundamental conflict between humanity's search for inherent meaning and the universe's silent indifference to that search; Camus argued we must nonetheless embrace life without illusory consolation.

13.4 Death and Interpersonal Existence

Existentialist philosophers treat death not as a topic to avoid but as a defining feature of human existence that gives life urgency and meaning — Martin Heidegger described authentic existence as living with full awareness

of one's own 'being-toward-death.' Thanatology, the broader philosophical and scientific study of death and its meaning for human life, engages closely with these questions.

Sartre also emphasized 'the Other' — other people who, through their own perspective and gaze, affect and help constitute one's own self-identity — making interpersonal relationships a central and unavoidable part of authentic existence, requiring mutual recognition and responsibility rather than isolation.

13.5 Philosophy, Human Rights, and the Environment

Beyond its classical themes, contemporary philosophy actively engages practical human concerns: offering rational justification and critical reflection in debates over human rights (rather than merely religious backing), and, in environmental philosophy, questioning anthropocentrism — the human-centered view that only humans possess inherent moral worth — in favor of a more ethically inclusive relationship between humanity and the natural world.

Chapter Summary

Putting the whole course together:

- Philosophy is the rational, critical pursuit of fundamental questions (Unit 1), which unfolded historically from the ancient Greeks through medieval, modern, and contemporary thought (Unit 2), producing recurring theoretical themes like rationalism, empiricism, and pragmatism (Unit 3).
- Its two great theoretical branches are metaphysics — what is real, including substance and universals (Unit 4) — and epistemology — what we can know, and how justified we can be in claiming to know it (Unit 5).
- Logic (Unit 6), critical thinking and fallacies (Unit 7), syllogistic argument forms (Unit 8), and symbolic logic (Unit 9) supply the formal tools philosophy uses to reason correctly about all of the above.
- Philosophy's applied branches extend these tools to practical life: ethics (Unit 10) asks how we should act; philosophy of religion (Unit 11) asks about God, the soul, and ultimate meaning; political philosophy (Unit 12) asks how political authority and justice are justified; and existentialist philosophy (Unit 13) asks what it means to exist freely, authentically, and responsibly as a finite human being among others.